2501-PYL1003 (Mechanics & Special Theory of Relativity)

Practice Mid-Semester Examination — Paper 4 (Unofficial, Instructor-Style)

Full Marks: 60 Time: 2 hours Set: Practice-4

Instructions

An MCQ can also be an MSQ (select all correct options). Justify every choice — no marks for a bare letter.
Show all intermediate steps and reasoning.
State any approximation or rounding explicitly, with justification.
Numerical answers should be rounded to two decimal places unless stated otherwise.

1. A quantity Λ (the "cosmological constant") has dimension L^{-2}. It is proposed that Λ, c (speed of light), and H_0 (Hubble constant, dimension T^{-1}) can be combined with G (gravitational constant) to form a dimensionless density parameter Ω_Λ via

$$\Omega_\Lambda = \frac{\Lambda c^a}{H_0^b}$$

Determine the exponents a, b using dimensional analysis alone (i.e., find a, b such that the right side is dimensionless). **[5]**

2 (MSQ). A student claims the following equations are dimensionally valid, where x is position (L), t is time (T), v_0, a_0 are constants: a) $x = v_0 t + \frac{1}{2} a_0 t^3$, with a_0 having dimension LT^{-3} b) $x = v_0 t\, e^{a_0 t}$, with a_0 having dimension T^{-1} c) $v_0^2 = 2a_0 x$, with a_0 having the usual dimension LT^{-2} d) $x = a_0 \sin(v_0 t)$, with a_0 having dimension L and v_0 having dimension T^{-1} Which statement(s) about dimensional consistency is/are correct? **[4]**

3. A particle undergoes motion such that its acceleration is proportional to the **cube** of its velocity: $a = -\kappa v^3$ (for $v > 0$, motion along a line, $\kappa > 0$ constant), starting with initial speed v_0 at $t = 0, x = 0$. (i) Solve for $v(t)$. (ii) Solve for $x(t)$. (iii) Does the particle come to rest in finite time? Does it travel a finite total distance? Justify both answers separately (they may differ!). **[3+3+3 = 9]**

4. A block of mass 5 kg is on a horizontal surface with a **position-dependent** coefficient of kinetic friction $\mu(x) = 0.1 + 0.02x$ (x in metres, block moving in $+x$ direction, $g = 10\ \mathrm{m/sec^2}$). The block starts at $x = 0$ with speed $v_0 = 8$ m/sec and decelerates solely due to friction (no other horizontal force). (i) Using the work-energy theorem in differential form ($mv\,dv = -\mu(x) mg\, dx$), set up and solve the resulting differential equation to find v^2 as a function of x. (ii) Find the distance at which the block comes to rest (solve the resulting quadratic in x; discard unphysical roots). **[4+3 = 7]**

5 (MSQ). A point has spherical polar coordinates $r = 10, \theta = 143.13^\circ, \phi = 210^\circ$ (angles rounded, $z = r\cos\theta$, ϕ from $+x$-axis). Which of the following is/are correct about its Cartesian coordinates (two decimal places)? a) $z = -8.00$ b) $z = 8.00$ c) $x \approx -5.20, y \approx -3.00$ d) The cylindrical radius $\rho = 6.00$ **[6]**

6. A particle moves so that in spherical polar coordinates, $r(t) = r_0$ (constant), $\theta(t) = \theta_0$ (constant), and $\phi(t) = \omega t$ (i.e., uniform circular motion on a "latitude circle" of a sphere of radius r_0). (i) Using $\vec{v} = \dot{r}\,\hat{r} + r\dot{\theta}\,\hat{\theta} + r\sin\theta\,\dot{\phi}\,\hat{\phi}$, write $\vec{v}(t)$ for this motion. (ii) Using the general spherical-coordinate acceleration formula

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\,\dot{\phi}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\sin\theta\cos\theta\,\dot{\phi}^2)\hat{\theta} + (r\sin\theta\,\ddot{\phi} + 2\dot{r}\sin\theta\,\dot{\phi} + 2r\cos\theta\,\dot{\theta}\dot{\phi})\hat{\phi}$$

simplify $\vec{a}$ for this specific motion, and verify that its magnitude equals $\rho_0\omega^2$ where $\rho_0 = r_0\sin\theta_0$ is the radius of the latitude circle (i.e., confirm it reduces to ordinary centripetal acceleration for circular motion). **[3+6 = 9]**

7. Halley's Comet has a perihelion distance of roughly $r_p = 0.586$ AU and an orbital eccentricity of $\epsilon = 0.967$. (i) Find the semi-major axis a and the aphelion distance r_a (in AU). (ii) Using Kepler's third law in the form $T^2 \propto a^3$ (with T in years and a in AU for objects orbiting the Sun, taking the Sun-centred proportionality constant to be 1 in these units), find the orbital period T. (iii) Find the ratio of the comet's speed at perihelion to its speed at aphelion. **[3+2+2 = 7]**

8. A particle of mass m moves under a central force with potential energy $V(r) = \dfrac{A}{r^2} - \dfrac{B}{r}$ $(A, B > 0)$, with angular momentum L. (i) Write the full effective potential $V_{\text{eff}}(r)$ (combining $V(r)$ with the centrifugal term), and find the equilibrium radius r_0 for a circular orbit in terms of A, B, L, m (state clearly when r_0 is real/positive, i.e., what condition on L^2/m relative to A is required). (ii) Determine whether the circular orbit at r_0 is stable, by examining the sign of $V''_{\text{eff}}(r_0)$. **[5+4 = 9]**

9. A pendulum of length $\ell = 2$ m is set swinging in a vertical plane at a location on Earth with latitude $\lambda = 45^\circ$ N (this is the basic setup behind the Foucault pendulum). Using the result that, due to the Coriolis force, the plane of oscillation of a Foucault pendulum precesses at angular rate $\Omega_{\text{prec}} = \Omega\sin\lambda$ (where Ω is Earth's rotation rate): (i) Derive (do not merely quote) this precession rate by considering the horizontal projection of the Coriolis force on a pendulum bob moving with horizontal velocity $\vec{v}_h$, and arguing (via the small-oscillation, slow-precession approximation) that the bob's plane of swing rotates at rate $\Omega\sin\lambda$. [A clear, physically reasoned derivation — e.g. via the horizontal component of $\vec{\Omega}$ acting like a local effective rotation rate $\Omega\sin\lambda$ about the local vertical — is acceptable; full rigor is not required, but the key physical step must be justified.] (ii) Find the time required for the plane of swing to precess by 90° at this latitude. **[6+2 = 8]**

10. An elevator cabin accelerates **upward** with constant acceleration $A_0 = 2\ \text{m/sec}^2$ in an inertial (non-rotating) frame — this is a **non-inertial linearly-accelerating frame**, not a rotating one. Inside, a simple pendulum of length $\ell = 1$ m hangs from the ceiling. (i) Write down the effective gravitational field $\vec{g}_{\text{eff}}$

experienced by objects inside the elevator (magnitude and direction) in terms of g and A_0. (ii) Find the period of small oscillations of the pendulum inside the accelerating elevator ($g = 9.8\ \mathrm{m/sec^2}$). **[3+3 = 6]**

11. A jet of water of cross-sectional area $S = 4 \times 10^{-4}\ \mathrm{m^2}$ and speed $v_j = 15$ m/sec (density $\rho_w = 1000\ \mathrm{kg/m^3}$) strikes a **stationary, flat** vertical plate perpendicularly and, after striking, spreads out sideways along the plate (so the water leaves with zero velocity component along the original jet direction — it does not bounce back). (i) Using the variable-mass / momentum-flux approach (rate of mass striking the plate $\times$ change in velocity component along the jet direction), find the force exerted by the jet on the plate. (ii) If the plate is now allowed to move away from the jet at constant speed $v_p = 5$ m/sec (in the same direction the jet is pushing it), find the new force on the plate, accounting correctly for the **relative** speed of water striking the (now receding) plate and the correspondingly reduced mass flux hitting it per unit time. **[4+5 = 9]**

12. A conveyor belt moving horizontally at constant speed v_{belt} carries sand that is loaded onto it at a constant rate σ (kg/sec) at one end, from a hopper with negligible horizontal velocity (as in Paper 2 Q12, but now asked differently): (i) If instead the belt speed is **not** held constant, but the belt (of mass M_{belt}, treat as fixed, non-accelerating support structure — only consider the sand's dynamics and the horizontal driving force) is driven by a motor delivering **constant power** P_0 (rather than constant force), set up the differential equation governing $v_{belt}(t)$ assuming at $t = 0$ the belt (and the sand layer already on it) has speed $v_{belt}(0) = v_1 > 0$ and total sand mass on the belt at time t is $m(t) = m_0 + \sigma t$. [You do not need to fully solve the resulting ODE in closed form — setting it up correctly, with correct identification of the thrust/loading term and the power constraint, is sufficient. If you can solve it or reduce it to quadrature, do so for extra credit.] **[8]**

13. A block of mass $m_1 = 6$ kg is connected via a light string over a frictionless pulley to a second block $m_2 = 4$ kg. Block m_1 sits on a horizontal table, and the string from m_1 passes over the pulley fixed at the table's edge and then connects to m_2, which hangs freely — **but** the entire table (and pulley) is itself mounted on a cart that is accelerating horizontally (in the direction along the string on the table, toward the pulley) at constant $a_{cart} = 3\ \mathrm{m/sec^2}$ relative to the ground. The table is frictionless. Analyze the system in the **non-inertial cart frame** (introduce the pseudo-force on each block due to a_{cart}). (i) Write the equations of motion for m_1 (horizontal, in the cart frame) and m_2 (vertical — note m_2 also experiences a horizontal pseudo-force, but since it hangs vertically and swings are neglected/it's constrained to move purely vertically relative to the cart via the string+pulley geometry, focus on the string-direction equation) in the cart frame, including the pseudo-force terms. (ii) Solve for the acceleration of the blocks **relative to the cart**, and the tension in the string ($g = 10\ \mathrm{m/sec^2}$). **[4+5 = 9]**

14 (MSQ). A uniform plank of mass M and length L rests horizontally on two supports: one at its left end, and one at distance d from the left end (with $d < L$), i.e., the right portion of the plank (length $L - d$) overhangs the second support. A small mass m is placed at the very right end of the plank (at the overhanging tip). Which of the following statement(s) is/are correct regarding whether the plank tips over (rotates about the second support)? a) The plank is on the verge of tipping when the net torque about the second support (from gravity on the plank and on m) is zero b) Tipping occurs if $m(L - d) > M\left(d - \dfrac{L}{2}\right)$, assuming $d > L/2$ c) If $d = L/2$ exactly (second support at the plank's center), the plank tips for **any** nonzero m placed at the right tip d) The first support (at the left end) can only push up on the plank, never pull down — this matters for determining when the plank "lifts off" the left support as tipping begins **[6]**

15. A particle of mass m moves under a central force $F(r) = -\dfrac{k}{r^2} + \dfrac{h}{r^3}$ (a superposition of inverse-square attraction, $k > 0$, and an additional inverse-cube term, h can be of either sign), with angular momentum L. (i) Show that this problem can be mapped onto an **effective** inverse-square problem with a **modified** angular momentum L', by absorbing the h/r^3 term into the centrifugal part of the effective potential. Find L'^2 in terms of L^2, m, h (state the condition on h, relative to L^2/m, for L'^2 to remain positive). (ii) Given this mapping, state (without full re-derivation, by analogy to the standard Kepler problem) the form of the orbit $r(\theta)$ for this force law when $L'^2 > 0$, and explain qualitatively how the orbit's **precession** (the angle swept between successive perihelia, generally $\neq 2\pi$) arises from the relationship between L' and L. **[5+4 = 9]**

End of Paper 4. Total: 60 marks.